

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2019

Subject: 341151 Electric Circuits I

Section: 5-6

Date: 17 August 2019

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 12 pages (including this page) and they are worth 65 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (15 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. 1,000,000 Hz is 10 MHz.
2. The prefix that corresponds to 10^{12} is Giga.
3. Every source of voltage is created by simply making a separation of positive and negative charges.
4. A potential difference of 1 volt (V) exists between two points if 1 joule (J) of energy is exchanged in moving 1 coulomb (C) of charge between the two points.
5. Generally, notation for dc voltage sources is E and voltage across resistor is V.
6. Conventional current is in the opposite direction of the electron current.
7. Primary type of battery is not rechargeable.
8. From Coulomb's law $F = k \frac{Q_1 Q_2}{r^2}$, where k denotes a constant of 9×10^9 .
9. Kirchhoff's current law requires that total current input to a system must equal to the total current output from the system.
10. $\eta = \frac{P_o}{P_{in}}$ denotes the efficiency of a system.
11. Following the Ohm's law, the power loss at resistor is $P=VI$.
12. The kilowatthour meter is used to measure only electrical current.
13. The circuit breaker is designed to protect short circuit.
14. A 12 V battery will have longer battery life than a 6 V battery.
15. Internal resistance of the voltmeter is in Mega ohm range, thus it is not affected to the circuit.

Part 2 (30 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. The voltage divider rule states that for two series _____ of equal value, the voltage will divide equally.
(a) elements (b) resistors
(c) circuits (d) branches

2. Four 3-volt batteries connected in parallel will result in which of the following voltage?
(a) 3 volts (b) 6 volts
(c) 12 volts (d) 48 volts

3. Find voltage between two points if the energy of 10 J is required to move a charge of 10 C between two points.
(a) 100 V (b) 10 V
(c) 1 V (d) 0.1 V

4. Evaluate the following expression: $(15 \times 10^{-6}) / (5 \times 10^{-9}) =$
(a) 3×10^{-3} (b) 3×10^3
(c) 0.03 (d) 300

5. A voltmeter is always placed _____ an element to measure the voltage.
(a) in parallel with (b) after
(c) in series with (d) on top of

6. Four 12-volt batteries are connected together by series configuration. What is a total voltage?
(a) 3 volts (b) 6 volts
(c) 12 volts (d) 48 volts

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7. See Figure 1. The total resistance in this circuit is

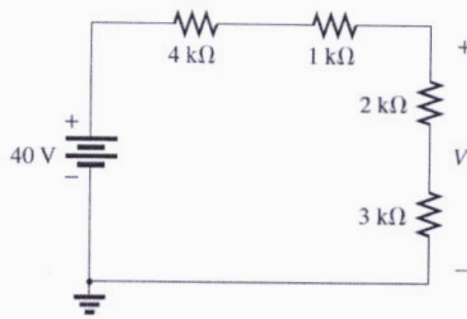


Figure 1.

- (a) $1 \text{ k}\Omega$ (b) $2 \text{ k}\Omega$
 (c) $7 \text{ k}\Omega$ (d) $10 \text{ k}\Omega$

8. See Figure 1. The total current flowing from the battery is

- (a) 1 mA (b) 10 mA
 (c) 1.43 mA (d) 4 mA

9. See Figure 1. The total power dissipated by the $3 \text{ k}\Omega$ resistor is

- (a) 12 mW (b) 48 mW
 (c) 7 mW (d) 10 mW

10. Find the energy to move charge of 10 C between two points if the voltage is 5 V.

- (a) 500 J (b) 50 J
 (c) 5 J (d) 0.5 J

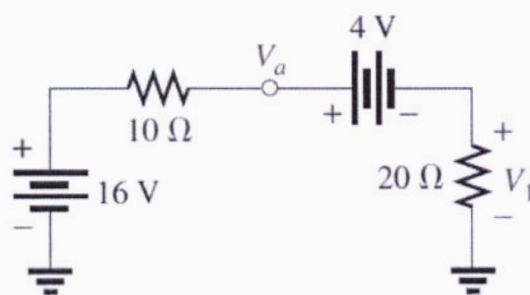


Figure 2.

11. See Figure 2. The total resistance in this circuit is

- (a) $10\ \Omega$ (b) $20\ \Omega$
 (c) $30\ \Omega$ (d) $40\ \Omega$

12. See Figure 2. The total current is

- (a) $+500\text{ mA}$ (b) -500 mA
 (c) $+2.5\text{ A}$ (d) -2.5 A

13. A circuit has a source voltage of 15 volts and two resistors with a total resistance of 4000 ohms. If R_1 has a potential of 9.375 volts, what is the value of R_2 ?

- (a) 500 ohms (b) 1000 ohms
 (c) 1500 ohms (d) 2000 ohms

14. See Figure 3. What is the total current I ?

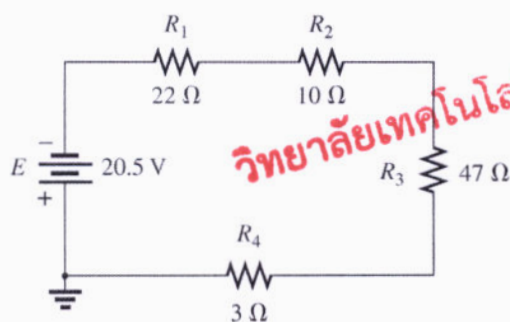


Figure 3.

- (a) 2.5 mA (b) 0.25 mA
 (c) 2.5 A (d) 0.25 A

15. See Figure 3. The total power dissipated by the $3\ \Omega$ resistor is

- (a) 0.75 W (b) 2.5 W
 (c) 0.33 mW (d) 1.75 mW

16. The attraction force between two charges which can be determined by_____.

- (a) Coulomb's law (b) Newton's law
(c) Kirchhoff's current law (d) Ohm's Law

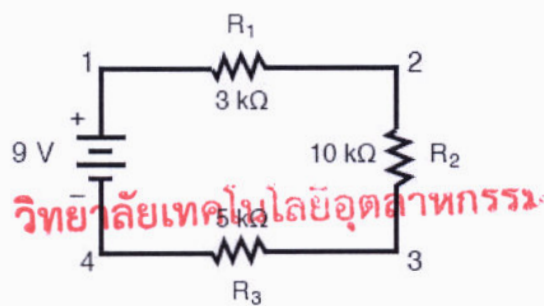
17. How much charge passes through a battery of 4.5 V if the energy expended is 60 J?

- (a) 13.5 C (b) 13.3 C (c) 10.6 C (d) 11.4 C

18. What is the total cost of using 1,000 W air conditioner for 12 h at a rate of 10 Baht per kilowatthour?

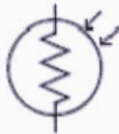
- (a) 160 Baht (b) 120 Baht (c) 12.0 Baht (d) 240 Baht

19. Following the circuit below, find the current I.

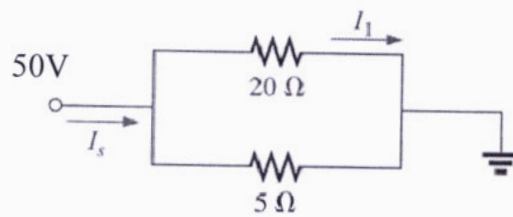


- (a) 1.5 mA (b) 0.5 mA (c) 1.8 mA (d) 18.5 mA

20. What is the symbol of potentiometer?

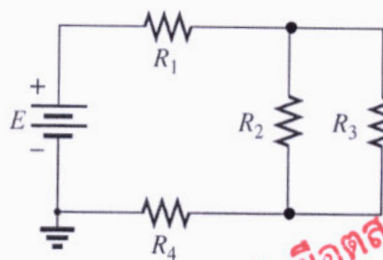
- (a)  (b)  (c)  (d) 

21. Find the value for the unknown current I_1 .



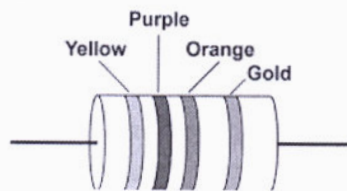
- (a) -7.5 A (b) +7.5 A
(c) -1.2 A (d) +2.5 A

22. Find the value of R_1 . Given $R_2=10\ \Omega$, $R_3=10\ \Omega$, $R_4=5\ \Omega$ and $R_T=20\ \Omega$.



- (a) $9\ \Omega$ (b) $5\ \Omega$ (c) $10\ \Omega$ (d) $20\ \Omega$

23. Find the value of R in picture below.

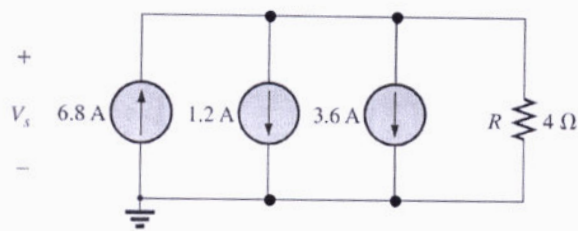


- (a) $47\ \Omega$ (b) $4.7\ \text{k}\Omega$ (c) $47\ \text{k}\Omega$ (d) $4.7\ \Omega$

24. Find the dissipated power of R colored green, red, blue, gold and gold, where the current is 1 A.

- (a) 526 W (b) 52.6 W (c) 5.26 W (d) 0.526 W

25. Determine the voltage V_R



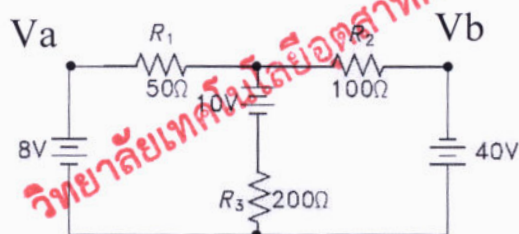
- (a) 8 V (b) -8 V (c) 46.4 V (d) -46.4 V

26. What is the meaning of 5 W resistor?



- (a) 5 W of dissipated power.
 (b) 5 W of maximum power handling.
 (c) 5 W power is converted to the light.
 (d) 5 W of minimum power handling.

27. Find the value of V_{ab} .

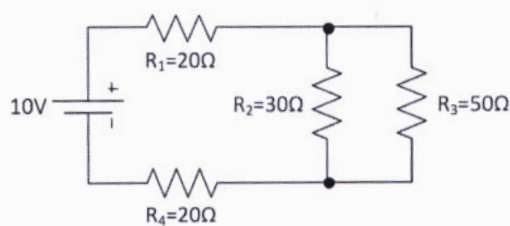


- (a) 48 Volts (b) -48 Volts (c) 32 Volts (d) -32 Volts

28. How large is dissipated power (W) of a 5Ω resistor connected across a 10 V battery?

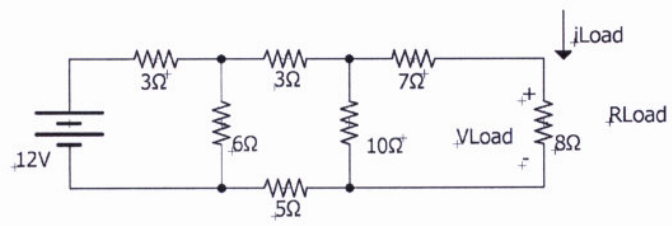
- (a) 20 W (b) 2 W (c) 10 W (d) 1 W

29. Calculate R_T



- (a) 38.75Ω (b) 58.75Ω (c) 70Ω (d) 90Ω

30. Following the circuit below, find total resistance R_T .



(a) $9.2\ \Omega$

(b) $7.2\ \Omega$

(c) $10.2\ \Omega$

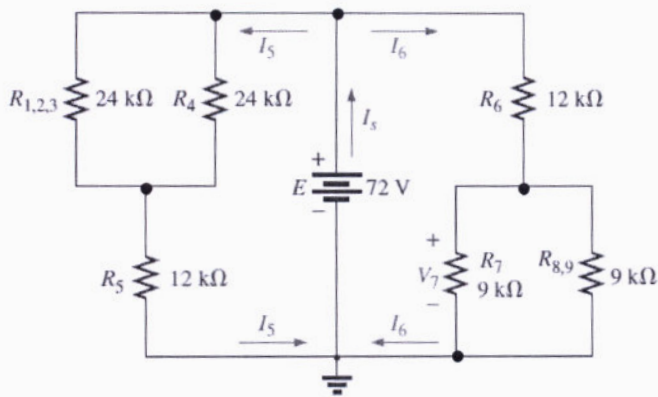
(d) $20\ \Omega$

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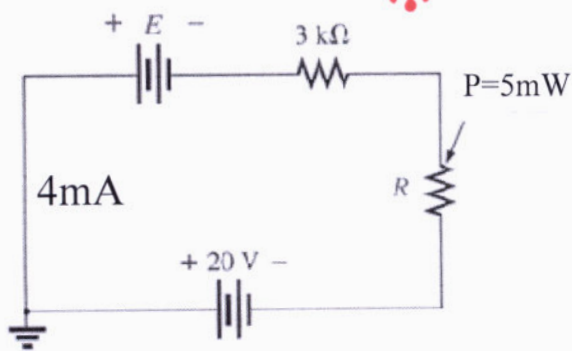
Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

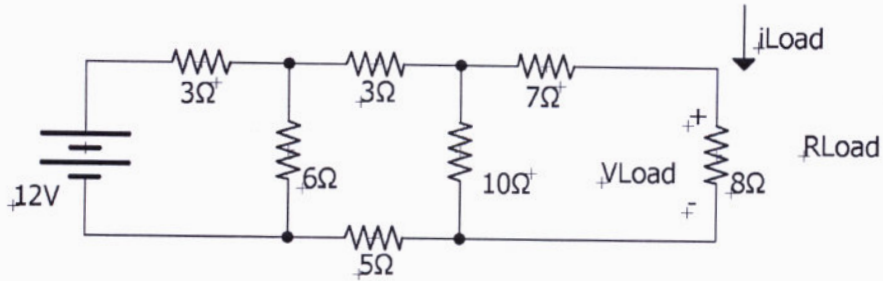
1. From Figure below, calculate R_T and I_S . (5 points)



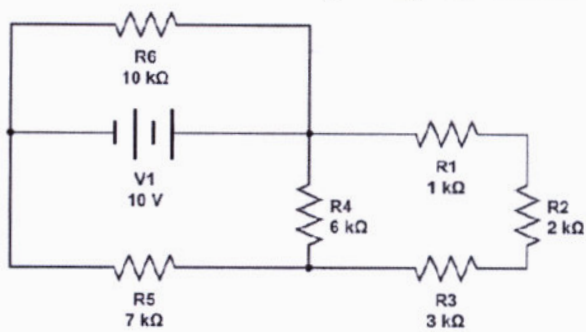
2. Find the unknown voltage source E and resistor R for the networks below. Also indicate the direction of the resulting current. (5 points)



3. Determine I_{Load} and V_{Load} from the following circuit. (5 points)



4. Find total current I_T , voltage V_{R1} and V_{R4} . (5 points)



Instructors: Assoc. Prof. Dr. Titipong Lertwiriaprapa

Asst. Prof. Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 341151 Electric Circuits I

Part 1

	True	False
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Part 2

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